



TEST CODE **02120020**

MAY/JUNE 2006

FORM TP 2006206

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCE USE

PAPER 02

2 hours 30 minutes

09 MAY 2006 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **NINE** questions.
2. Answer **SIX** questions.
3. Write your answers in the answer booklet provided.
4. The use of a non-programmable calculator is allowed.

MODULE 1

Answer ANY TWO questions.

1. (a) Explain the meaning of
- (i) range of tolerance [4 marks]
 - (ii) limiting factors. [4 marks]
- (b) State THREE limiting factors for an aquatic ecosystem. [3 marks]
- (c) Outline how EACH of the factors in (b) above impacts on the population of organisms in an aquatic ecosystem. [9 marks]

Total 20 marks

2. Figure 1 represents a pyramid of energy for a mango tree.

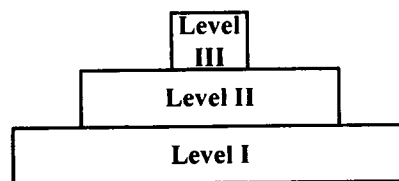


Figure 1. Pyramid of energy

- (a) Identify the categories of organisms at EACH level. [3 marks]
- (b) Account for the shape of the diagram. [5 marks]
- (c) Draw a diagram showing the pyramid of numbers for the mango tree. [3 marks]
- (d) Account for any difference between the shape of the pyramid of energy shown in Figure 1 and the pyramid of numbers drawn in (c) above. [9 marks]

Total 20 marks

3. "Natural ecosystems are vital to the existence of human beings."

Justify the statement above. Include FIVE reasons in your answer.

Total 20 marks

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MODULE 2

Answer ANY TWO questions.

4. Figure 2 shows the size of the population of a Caribbean country during the period 1840 to 2000.

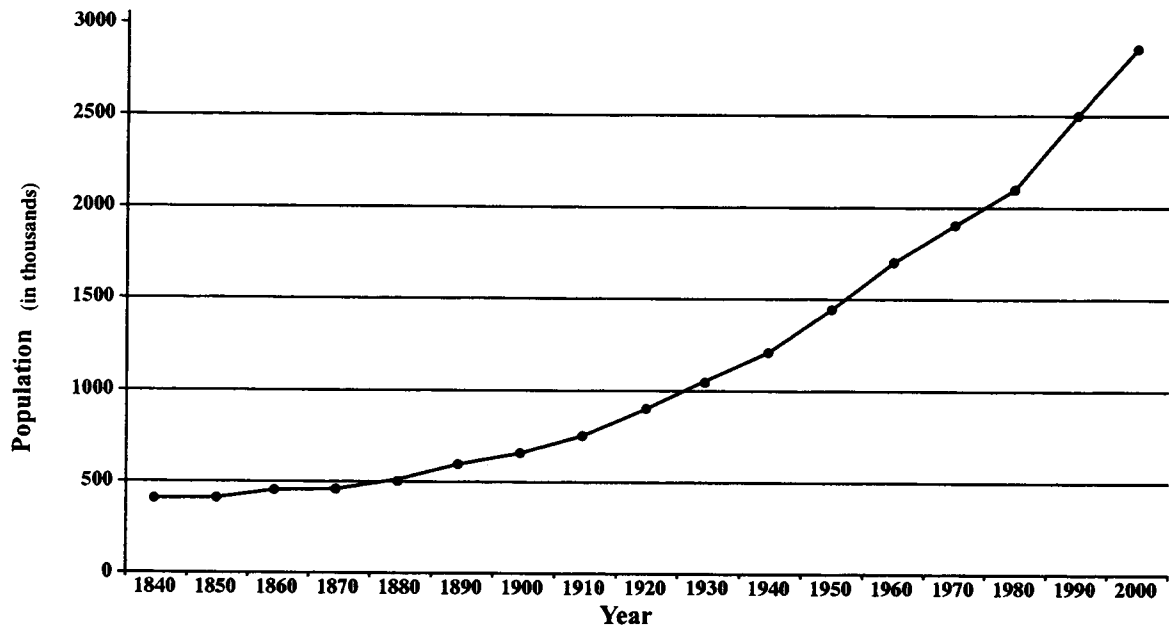


Figure 2. Population of a Caribbean country

- (a) Describe, in terms of the birth and the death rate, how the rate of growth of the population depicted in Figure 2 changed between 1840 and 2000. [4 marks]
- (b) By considering at least THREE social parameters, suggest how changes in social conditions within the population during the period 1840 to 1880 could account for the changes in growth rate described in (a) above. [12 marks]
- (c) Outline ONE possible environmental impact of rapid changes in the population growth rate of a named Caribbean country. [4 marks]

Total 20 marks

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5. Figure 3 represents a map of the world which is divided into regions (1) to (8)

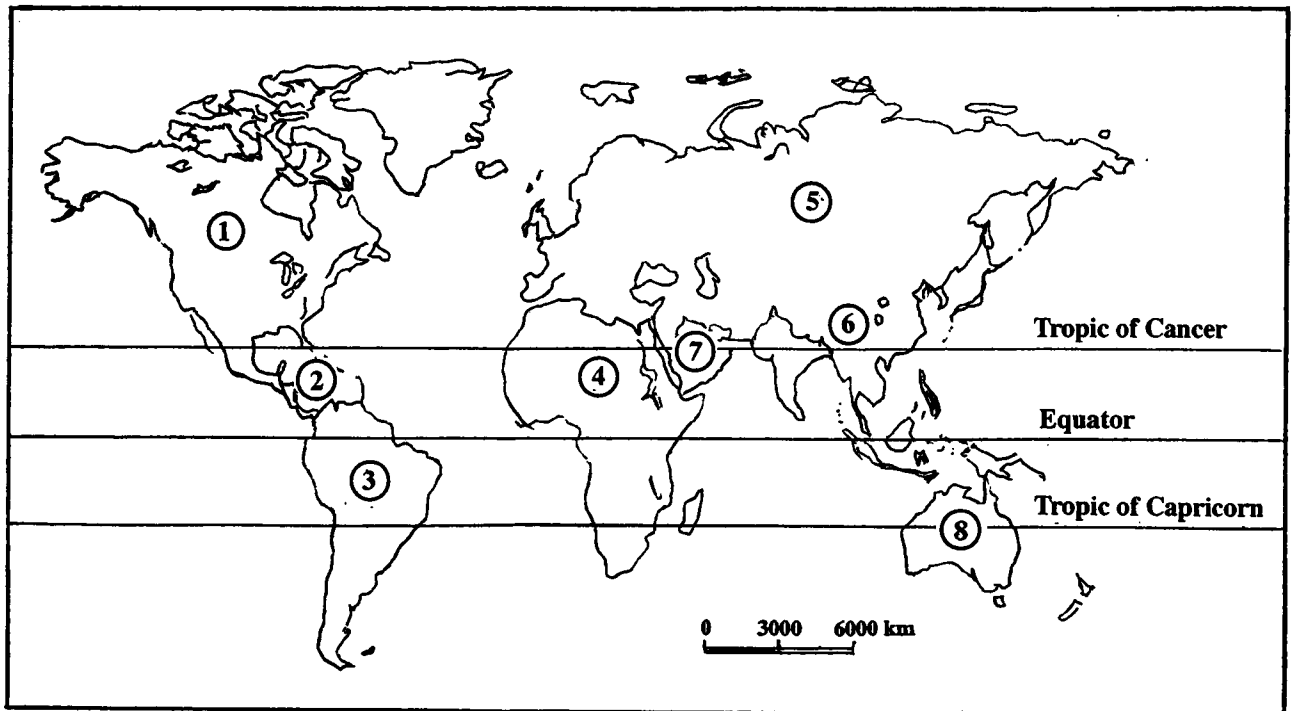


Figure 3. The distribution of the world's population

- (a) Place EACH of the regions, (1) to (8), into ONE of the following categories:
- (i) High per capita consumption
 - (ii) Low per capita consumption patterns [3 marks]
- (b) Evaluate the statement below.
- “High consumption patterns in developed countries have led to environmental problems in developing countries.” [10 marks]
- (c) Developing countries have low per capita consumption in comparison to developed countries. Does this mean that developing countries do not contribute significantly to the world's environmental problems? Justify your response. [7 marks]

Total 20 marks

6. (a) Outline the relationship between the population growth of a country and its ability to achieve sustainable development. [8 marks]
- (b) Explain how EACH of the following social indicators relating to women will impact on the sustainable development of **your** country.
- (i) Level of education
 - (ii) Economic status
 - (iii) Access to family planning services [12 marks]

Total 20 marks

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MODULE 3

Answer ANY TWO questions.

7. Describe EACH of the following environmental impacts associated with the exploitation of minerals in the Caribbean:

- (a) Physical conversion of land [5 marks]
- (b) Dust pollution [5 marks]
- (c) Discharge of process chemicals [5 marks]
- (d) Siltation [5 marks]

Total 20 marks

8. (a) Suggest FOUR types of reasons for natural resource conservation. [4 marks]

(b) Discuss the role of EACH of the following in natural resource conservation:

- (i) Rehabilitation [4 marks]
- (ii) Restoration [4 marks]
- (iii) Preservation [4 marks]
- (iv) In-situ and ex-situ conservation [4 marks]

Total 20 marks

9. (a) Describe FOUR environmental impacts associated with the exploitation of forest resources in the Caribbean. [16 marks]

(b) Suggest ONE recommendation for addressing EACH impact identified in (a) above. [4 marks]

Total 20 marks

END OF TEST